Percolation by Simulation: A Problem Set

Companion to "Percolation by Example"

October 2026

This problem set turns Examples 1–3 (and, optionally, Example 5) of the main text into simulation exercises. Each problem states a goal, what to implement, what to plot, and *checkpoints*: analytic numbers your code should reproduce before you trust it on cases where no analytic answer exists. Problems marked $\star$ are warm-ups, $\star\star$ are core, and $\star\star\star$ are open-ended. No solutions or code are given.

General rules for all problems.

1. **Fix and record the random seed** for every run, so results are reproducible.
2. **Report error bars.** For any averaged quantity, report the standard error over independent realisations, $\sigma/\sqrt{M}$.
3. **Test before you measure.** Every new routine is first run on tiny, hand-made configurations whose answer you can count by eye.
4. **Vary the system size.** Any claimed threshold must be shown for at least three system sizes.
5. **Write your own cluster labelling** in Part 0; afterwards you may compare against a library routine as a check.

0 Toolkit

Problem 0.1 Union–find cluster labelling $\star$

Implement a *union–find* (disjoint-set) data structure with two operations: `find(i)` returns the representative of the set containing i, and `union(i,j)` merges two sets. Add *path compression* and *union by size*, and keep track of the size of each set.

Tasks.

(a) Use it to label clusters of (i) occupied sites on a 1D chain, (ii) occupied sites on an $L \times L$ square lattice (site percolation), (iii) open bonds on an $L \times L$ square lattice (bond percolation), and (iv) an arbitrary graph given as an edge list.

(b) Construct at least five 5×5 test grids by hand (all empty, all full, a checkerboard, a single snake-like cluster, two clusters touching only diagonally) and verify the cluster sizes.

(c) Measure the run time for $L = 64, 128, \ldots, 2048$ at $p = 0.6$. How does it scale with the number of sites?

Think about it. Why is depth-first search with recursion risky for large clusters near p_c? What does path compression buy you?

Problem 0.2 Spanning and the largest cluster ⋆

Write two diagnostics for a labelled lattice configuration: (i) the size of the largest cluster, divided by the number of sites; (ii) a boolean "spans", true if one cluster touches both the left and the right boundary.

Hint (no code). Two extra "virtual" sites, one attached to every site in the left column and one to every site in the right column, turn spanning into a single `find` comparison.

A The one-dimensional chain (Example 1)

Problem A.1 Cluster-size distribution ⋆

Simulate a chain of L sites, each occupied with probability p.

Tasks.

(a) Measure n_s, the number of clusters of size s *per site*, for $p = 0.5, 0.8, 0.95$ and $L = 10^5$, and plot it on a semi-log axis against $(1-p)^2 p^s$.

(b) Do this once with open and once with periodic boundaries. Where do they differ, and why does the difference grow as $p \to 1$?

(c) Verify numerically that $\sum_s s\, n_s = p$.

Checkpoint. At $p = 0.5$, $n_1 = 0.125$ and $n_2 = 0.0625$.

Problem A.2 Two kinds of "mean cluster size" ⋆⋆

This problem makes the distinction between cluster-weighted and site-weighted averages concrete.

Tasks.

(a) From your n_s, compute the cluster average $\langle s \rangle_{\text{cluster}} = \sum s\, n_s / \sum n_s$ and the site average $\chi = \sum s^2 n_s / \sum s\, n_s$.

(b) Estimate χ a second, independent way: pick 10^4 occupied sites uniformly at random and average the size of the cluster each belongs to. Do the two estimates agree within error bars?

(c) Plot both averages against p together with $1/(1-p)$ and $(1+p)/(1-p)$.

Checkpoint. At $p = 0.5$: $\langle s \rangle_{\text{cluster}} = 2$, $\chi = 3$. At $p = 0.9$: $\langle s \rangle_{\text{cluster}} = 10$, $\chi = 19$.

Think about it. Repeat (b) but pick a random *cluster* first and then a random site in it. Which average do you get, and why?

Problem A.3 The threshold of a finite chain ⋆⋆

In 1D, $p_c = 1$ only for an infinite chain. For a chain of length L with open boundaries, "spanning" means all L sites are occupied.

Tasks.

(a) Measure the spanning probability $\Pi_L(p)$ for $L = 10, 30, 100, 300$. Overlay the curves.

(b) Define an effective threshold $p_c(L)$ by $\Pi_L = 1/2$. Plot $1 - p_c(L)$ against L on log–log axes and fit a power law.

(c) Compare with what you can work out on paper for $\Pi_L(p)$.

Checkpoint. $p_c(10) \approx 0.933$, $p_c(100) \approx 0.9931$; the log–log slope is -1.

Think about it. The correlation length of the 1D chain is $\xi = -1/\ln p$. Show that $p_c(L)$ is where $\xi \sim L$. This is the simplest example of finite-size scaling.

Problem A.4 A correlated chain (a first non-ideal case) ⋆⋆

Real binding sites are rarely independent. Generate a chain as a two-state Markov process: a site is occupied with probability a if its left neighbour is occupied, and with probability b if its left neighbour is empty.

Tasks.

(a) Before simulating, predict on paper the overall occupied fraction and the distribution of cluster sizes. (What is the probability that a cluster that has reached size s grows to $s + 1$?)

(b) Simulate for $(a, b) = (0.9, 0.1)$, $(0.5, 0.5)$ and $(0.1, 0.9)$. Compare n_s and χ with your predictions and with an uncorrelated chain of the *same* occupied fraction.

Checkpoint. $(a, b) = (0.5, 0.5)$ must reduce to the uncorrelated chain with $p = 0.5$.

Think about it. Positive correlations ($a > b$) make large clusters much more likely at fixed density. Which biological mechanisms (cooperative binding, nucleation) produce $a > b$?

B Branching trees and the role of loops (Example 2)

Problem B.1 Growing clusters on a Bethe lattice ⋆⋆

You cannot store an infinite tree, but you never need to: grow the cluster of an occupied origin *on the fly*. Start from the origin with $z = 3$ open directions; each new site is occupied with probability p and, if occupied, opens $z - 1 = 2$ further directions. Stop at a maximum depth G (number of generations).

Tasks.

(a) Use a queue (breadth-first) rather than recursion. Record, for each realisation, the cluster size and whether it reached depth G.

(b) Measure the mean cluster size $\chi(p)$ for $p = 0.30, 0.35, \ldots, 0.48$ with $G = 15, 25, 40$ and compare with $(1 + p)/(1 - 2p)$.

(c) Near $p = 1/2$ the finite depth biases your result. Plot the relative error against G for $p = 0.45$ and 0.49. How large must G be?

(d) Histogram the cluster sizes at $p = 0.45$ and $p = 0.50$. What changes in the shape of the distribution?

Checkpoint. $\chi(0.40) = 7$, $\chi(0.45) = 14.5$, $\chi(0.49) = 74.5$ (as $G \to \infty$).

Hint (no code). Above $p = 1/2$ clusters can be infinite; cap the total number of sites as well as the depth, and record how often the cap is hit.

Problem B.2 The survival probability Q and the giant cluster ⋆⋆

Tasks.

(a) From the simulation of Problem B.1, estimate the probability $1 - Q_G$ that a single branch reaches depth G, for p between 0.4 and 0.9.

(b) Independently, iterate the map $Q_{g+1} = 1 - p + p\,Q_g^2$ numerically. Decide (and justify) the correct starting value Q_0 so that Q_g means "fails to reach depth g". Compare with (a).

(c) Plot $P_\infty = p\,(1 - Q^3)$ against p and check the slope 6 just above $p = 1/2$.

(d) At $p = 1/2$, how fast does Q_g approach 1 with g? Compare with $p = 0.4$ and $p = 0.6$. Why is convergence so slow exactly at threshold?

Checkpoint. $p = 0.6$: $Q = 2/3$, $P_\infty \approx 0.422$. $p = 0.75$: $Q = 1/3$, $P_\infty \approx 0.722$.

Problem B.3 General coordination number ⋆

Generalise Problems B.1–B.2 to coordination $z = 3, 4, 5, 6$.

Tasks. Locate p_c numerically (e.g. where χ diverges, or where Q first drops below 1) and compare with $1/(z - 1)$. Derive and test the general formula for χ.

Problem B.4 Sparse random graphs look like trees ⋆⋆

Build an Erdős–Rényi random graph with N nodes and mean degree c (each pair linked with probability $c/(N - 1)$), using your union–find on the edge list.

Tasks.

(a) Measure the giant-component fraction $S(c)$ for $N = 10^3, 10^4, 10^5$ and $0 < c < 4$.

(b) Derive, with the branching argument, the self-consistency condition $S = 1 - e^{-cS}$, solve it numerically, and overlay it.

(c) Count short loops (e.g. triangles) per node as N grows. Why does the tree argument become exact?

Checkpoint. $S(1.5) \approx 0.583$, $S(2) \approx 0.797$, $S(3) \approx 0.940$; threshold $c = 1$.

Problem B.5 Loops raise the threshold ⋆⋆

The Bethe lattice with $z = 4$ has $p_c = 1/3$. The square lattice also has four neighbours per site, but it has loops.

Tasks.

(a) Using Part 0, measure the spanning probability for *bond* percolation on $L \times L$ square lattices with $L = 16, 32, 64, 128$, and locate the point where the curves cross.

(b) Repeat for *site* percolation.

(c) Express the difference from the tree value as a statement about redundant links.

Checkpoint. Bond: $p_c = 1/2$ (exact). Site: $p_c \approx 0.5927$.

Think about it. This is the same lesson as Example 5 (relay): tree estimates are lower bounds in low dimensions.

C Multivalent molecules and gelation (Example 3)

Problem C.1 Mean-field gel point: numbers first ⋆

Tasks.

(a) Write a root-finder (bisection is fine) for α in $\alpha/(1-\alpha)^2 = 2Kfc$, and plot α against Kc for $f = 2, \ldots, 6$.

(b) Mark on each curve where α crosses $1/(f-1)$ and reproduce Fig. 2(b) of the main text.

Checkpoint. $Kc_{\mathrm{perc}} = 1/3,\ 3/32,\ 2/45,\ 5/192$ for $f = 3, 4, 5, 6$. At $Kc = 0.1$: $\alpha \approx 0.297$ ($f = 3$) and $\alpha \approx 0.382$ ($f = 5$).

Problem C.2 Random pairing of stickers ⋆⋆

This is the simulation behind Flory–Stockmayer theory. Take N molecules, each with f stickers (so Nf sticker "stubs"). For a given α, choose αNf stubs at random (make it even) and pair them uniformly at random. Each pair is a bond between the two molecules that own the stubs.

Tasks.

(a) Measure the largest-cluster fraction (in molecules) against α for $f = 3, 4, 6$ and $N = 10^3, 10^4, 10^5$. Locate α_c from the crossing of curves for different N.

(b) Derive on paper the gel fraction. Let u be the probability that a sticker does *not* lead to the infinite network; argue that $u = 1 - \alpha + \alpha u^{f-1}$ and that the gel fraction is $1 - u^f$. Overlay this prediction.

(c) Count "wasted" bonds: intramolecular bonds (both stubs on the same molecule) and repeated bonds between the same two molecules. How does their number scale with N?

(d) Measure the size distribution of the finite clusters (the sol) just below α_c. These are the pre-percolation clusters of the main text.

Checkpoint. $\alpha_c = 1/2,\ 1/3,\ 1/5$ for $f = 3, 4, 6$. Gel fraction ≈ 0.704 at $(f, \alpha) = (3, 0.6)$ and ≈ 0.854 at $(4, 0.5)$.

Problem C.3 Two-component (heterotypic) gels ⋆⋆

Many condensates form from two partners, e.g. SH3-domain and proline-rich-motif proteins, where an A sticker binds only a B sticker. Take N_A molecules of valence f_A and N_B of valence f_B.

Tasks.

(a) Every bond uses one A and one B sticker. Write the relation between α_A and α_B in terms of the sticker ratio $r = N_A f_A/(N_B f_B)$.

(b) Extend the branching argument (a path now alternates $A \to B \to A$) and derive the gel condition. Predict α_c for $f_A = f_B = 3$, $r = 1$.

(c) Simulate by random A–B pairing and verify. Then fix the total number of molecules and vary the ratio N_A/N_B. Is the gel easiest to form at stoichiometric balance? Is there a ratio beyond which no gel can form at all?

Checkpoint. $f_A = f_B = 3$, $r = 1$: $\alpha_A = \alpha_B = 1/2$ at the gel point.

Think about it. This "stoichiometry window" is a known feature of multivalent condensates. Why does an excess of one partner suppress networking?

Problem C.4 Gelation in space (no analytic answer) ⋆⋆⋆

Random pairing ignores space: any sticker can bind any other. Now place N molecules at random in a periodic box (2D or 3D) at number density ρ. A sticker may bind only to a free sticker on a *different* molecule within distance r_b.

Tasks.

(a) Implement a simple bonding rule: list all candidate molecule pairs within r_b, visit them in random order, and form a bond with probability q if both molecules still have a free sticker. (Think about whether to allow several bonds between the same pair.)

(b) For fixed $f = 3$, measure the achieved α and the largest-cluster fraction as functions of q, for several densities ρr_b^d. Locate α_c at each density.

(c) Compare $\alpha_c(\rho)$ with the mean-field $1/(f-1)$. At high density, is α_c higher or lower, and how does it correlate with the fraction of bonds that close loops (bonds between molecules already in the same cluster)?

(d) Is "spanning" (Problem 0.2) a good criterion here? Compare it with the largest-cluster fraction.

Think about it. Your bonding rule is not an equilibrium model. What would change in a Monte Carlo model where bonds form and break with a Boltzmann weight e^{ϵ/k_BT}? (Optional: implement it.)

D Optional: signal relay among cells (Example 5)

Problem D.1 Reproducing $n_c \approx 4.51$ ⋆⋆

Place cells as a Poisson process in an $L \times L$ box; link cells closer than R.

Tasks. Measure the left–right spanning probability against $n = \rho\pi R^2$ for $L/R = 10, 20, 40, 80$, and locate the crossing. Use a cell list (grid of side R) so that neighbour search is not $O(N^2)$.

Checkpoint. $n_c \approx 4.51$.

Problem D.2 Checking the overlap estimate ⋆

By Monte Carlo, sample a child at distance r from its parent with density $2r/R^2$ and estimate the mean fraction of the child's disc covered by the parent's disc. Then, in your relay simulation, record the actual number of *new* cells recruited per cell in each generation of a breadth-first spread. Compare with n (tree) and $0.41\,n$ (overlap estimate). Which generations deviate most, and why?

Checkpoint. Mean overlap fraction $1 - 3\sqrt{3}/(4\pi) \approx 0.587$.

Problem D.3 Unresponsive cells ⋆

Make each cell responsive with probability q. At fixed $n = 6, 9, 12$, measure the fraction of cells reached against q and test $q_c = n_c/n$.

Problem D.4 Beyond hard cutoffs: summed signals ⋆⋆⋆

In cAMP relay a cell responds to the *summed* signal of many neighbours. Replace the hard cutoff by a kernel: an excited cell j contributes $e^{-d_{ij}/\lambda}$ to cell i, and cell i fires when the total from all currently excited cells exceeds a threshold θ. Start from a small excited disc in the centre.

Tasks.

(a) Measure the final fraction of excited cells against density for several θ.

(b) For small θ (one excited neighbour suffices) you should recover ordinary continuum percolation with an effective range. For large θ (several neighbours needed) how does the transition change? Is it still continuous?

(c) Does the outcome now depend on the size of the initial excited region?

Think about it. Requiring several excited neighbours is the rule of *bootstrap percolation*, which can show abrupt transitions. This is one route from the idealised Example 5 toward realistic excitable media.

Checkpoint summary

Problem	Quantity	Value
A.1	n_1, n_2 at $p = 0.5$	0.125, 0.0625
A.2	$\langle s\rangle_{\text{cluster}}$, χ at $p = 0.5$; at $p = 0.9$	2, 3; 10, 19
A.3	$p_c(L)$ for $L = 10, 100$	0.933, 0.9931
B.1	χ at $p = 0.40$, 0.45, 0.49 ($z = 3$)	7, 14.5, 74.5
B.2	(Q, P_∞) at $p = 0.6$; at $p = 0.75$	(2/3, 0.422); (1/3, 0.722)
B.4	S at $c = 1.5$, 2, 3	0.583, 0.797, 0.940
B.5	square-lattice p_c (bond, site)	0.5, 0.5927
C.1	Kc_{perc} for $f = 3, 4, 5, 6$	0.333, 0.094, 0.044, 0.026
C.2	α_c for $f = 3, 4, 6$; gel fraction at $(3, 0.6)$, $(4, 0.5)$	1/2, 1/3, 1/5; 0.704, 0.854
C.3	α_c for $f_A = f_B = 3$, $r = 1$	1/2
D.1	continuum relay threshold	$n_c \approx 4.51$
D.2	mean parent–child overlap	0.587